

BEACONHOUSE NATIONAL UNIVERSITY



Name: Khadijah Zahoor

BNU ID: F2023-956

Total Marks: 20

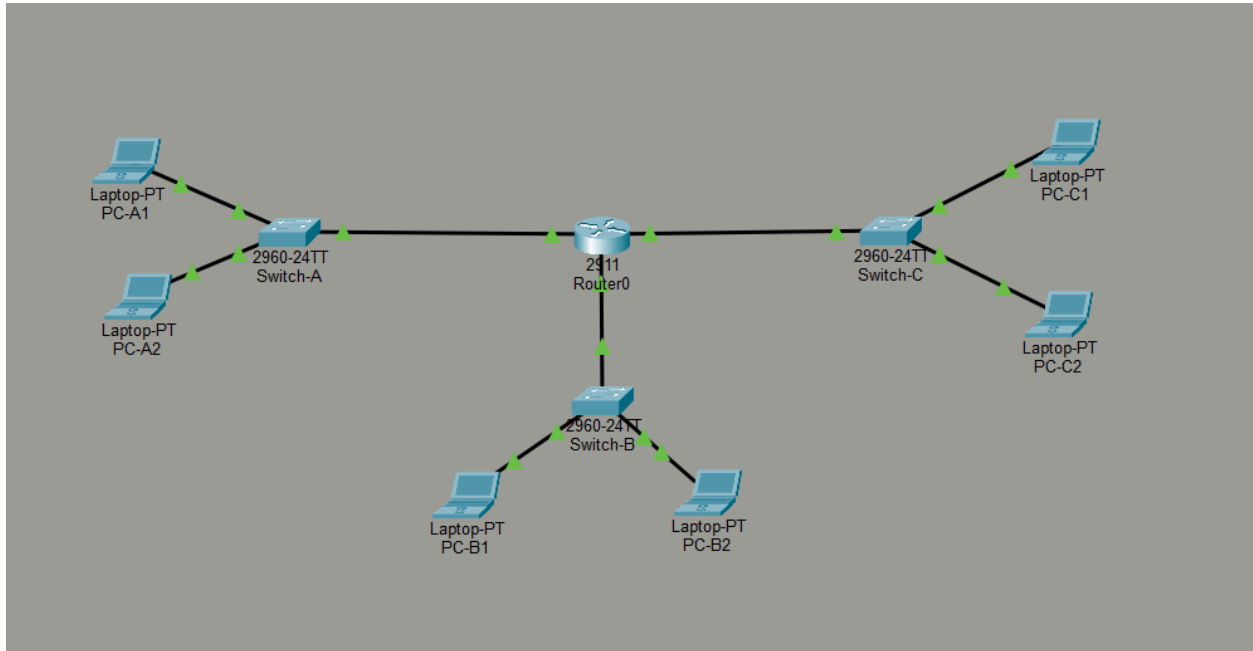
Due Date: 25-04-2026

Lab Assignment 1: Hybrid Network Topology Design Using Cisco Packet Tracer

Course Instructor	Ms. Shazia Rizwan
Lab Instructor	Sir Hasnain Ahmad
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Introduction

This assignment demonstrates the implementation of a hybrid network topology in Cisco Packet Tracer. The network consists of three star topology segments: IT Department, Admin Department, and Library. These segments are connected through a central router, which acts as the backbone. Static IP addresses were assigned manually, router interfaces were configured, and connectivity was verified using ping tests.



Assigning IP Addresses

PC-A1

The screenshot shows the configuration window for PC-A1. The 'Config' tab is active, and the 'IP Configuration' section is expanded. The interface is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 Address is set to 192.168.10.10, Subnet Mask to 255.255.255.0, Default Gateway to 0.0.0.0, and DNS Server to 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The IPv6 Address is empty, Link Local Address is FE80::209:7CFF:FE50:418B, Default Gateway is empty, and DNS Server is empty. Under '802.1X', the 'Use 802.1X Security' checkbox is checked, Authentication is set to MD5, and Username and Password fields are empty. A 'Top' button is at the bottom left.

Interface	FastEthernet0
IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☒ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::209:7CFF:FE50:418B
Default Gateway	
DNS Server	
802.1X	
☒ Use 802.1X Security	
Authentication	MD5
Username	
Password	

PC-A2

The screenshot shows the configuration window for PC-A2. The 'Config' tab is active, and the 'IP Configuration' section is expanded. The interface is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 Address is set to 192.168.10.11, Subnet Mask to 255.255.255.0, Default Gateway to 0.0.0.0, and DNS Server to 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The IPv6 Address is empty, Link Local Address is FE80::201:C9FF:FE57:CE05, Default Gateway is empty, and DNS Server is empty. Under '802.1X', the 'Use 802.1X Security' checkbox is checked, Authentication is set to MD5, and Username and Password fields are empty. A 'Top' button is at the bottom left.

Interface	FastEthernet0
IP Configuration	
☒ DHCP	☒ Static
IPv4 Address	192.168.10.11
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☒ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::201:C9FF:FE57:CE05
Default Gateway	
DNS Server	
802.1X	
☒ Use 802.1X Security	
Authentication	MD5
Username	
Password	

PC-B1

PC-B1

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address 192.168.20.10

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20C:85FF:FE03:D943

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication MD5

Username

Password

Top

PC-B2

PC-B2

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address 192.168.20.11

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20D:B0FF:FE54:9CC

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication MD5

Username

Password

Top

PC-C1

PC-C1

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCPStatic

IPv4 Address192.168.30.10

Subnet Mask255.255.255.0

Default Gateway0.0.0.0

DNS Server0.0.0.0

IPv6 Configuration

AutomaticStatic

IPv6 Address

Link Local AddressFE80::20A:F3FF:FE44:E486

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

PC-C2

PC-C2

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCPStatic

IPv4 Address192.168.30.11

Subnet Mask255.255.255.0

Default Gateway0.0.0.0

DNS Server0.0.0.0

IPv6 Configuration

AutomaticStatic

IPv6 Address

Link Local AddressFE80::260:3EFF:FE4D:8B7

Default Gateway

DNS Server

802.1X

Use 802.1X Security

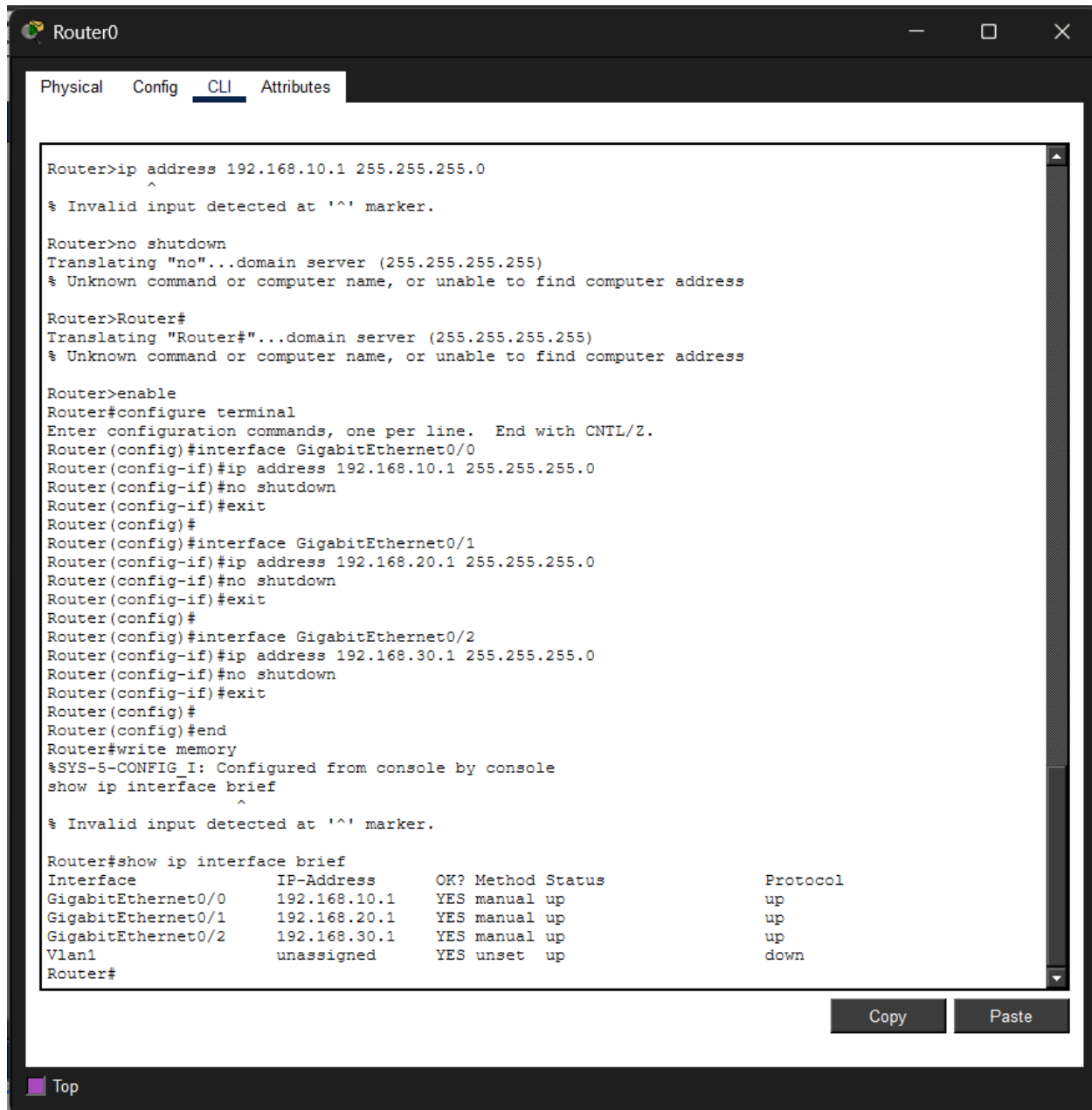
AuthenticationMD5

Username

Password

Top

Router0 CLI: show ip interface brief Output



The screenshot shows a terminal window titled "Router0" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying a series of commands and their outputs. The commands include setting IP addresses on three interfaces (GigabitEthernet0/0, 0/1, and 0/2), enabling them, and finally running the "show ip interface brief" command. The output of this command is a table showing the status of each interface.

```
Router>ip address 192.168.10.1 255.255.255.0
^
% Invalid input detected at '^' marker.

Router>no shutdown
Translating "no"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router>Router#
Translating "Router#"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#end
Router#write memory
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
^
% Invalid input detected at '^' marker.

Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.10.1    YES manual  up          up
GigabitEthernet0/1       192.168.20.1    YES manual  up          up
GigabitEthernet0/2       192.168.30.1    YES manual  up          up
Vlan1                    unassigned      YES unset   up          down
Router#
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

Gateway Connectivity Test

Gateway Test 1 : PC-A1 to Gateway

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Gateway Test 2 : PC-B1 to Gateway

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Gateway Test 3 : PC-C1 to Gateway

```
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time=1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping Test 1: PC-A1 to PC-A2 AND Ping Test 2: PC-B1 to PC-B2

```
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping Test 3: PC-C1 to PC-C2

```
C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Reply from 192.168.30.11: bytes=32 time<1ms TTL=128
Reply from 192.168.30.11: bytes=32 time=1ms TTL=128
Reply from 192.168.30.11: bytes=32 time=3ms TTL=128
Reply from 192.168.30.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

Ping Test 4: PC-A1 to PC-B1

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```


Ping Test 5 : PC-A1 to PC-C1

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping Test 6 : PC-B2 to PC-C2

```
C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Reply from 192.168.30.11: bytes=32 time<1ms TTL=127
Reply from 192.168.30.11: bytes=32 time<1ms TTL=127
Reply from 192.168.30.11: bytes=32 time<1ms TTL=127
Reply from 192.168.30.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Conclusion

The hybrid network topology was successfully created in Cisco Packet Tracer. Three star topology segments were connected through a central router acting as the backbone. Static IP addresses and default gateways were assigned to all PCs, and router interfaces were configured successfully. The ping tests verified both same-segment and cross-segment connectivity, confirming that the hybrid network is functioning properly.